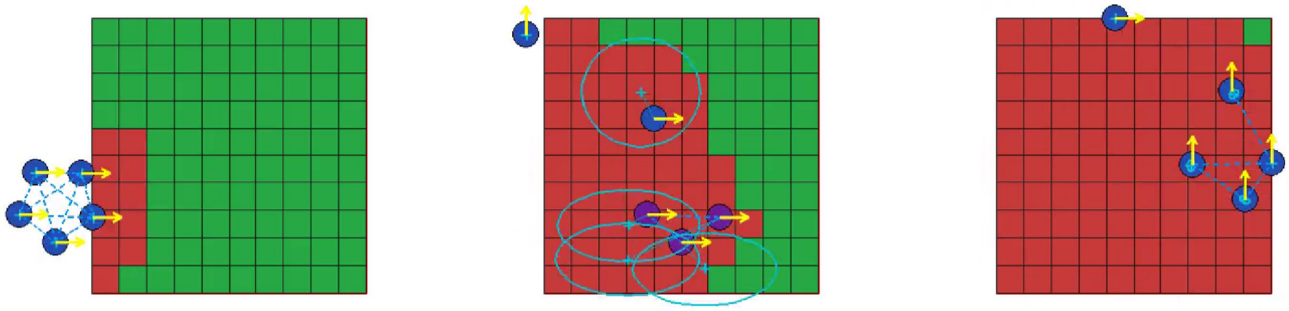
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Learning Uncertainty-Gated Coverage for GPS-Denied Multi-Robot Teams

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(a) $t = 4$ (b) $t = 36$ (c) $t = 64$

Fig. 1. Greedy MAPPO trial under moderate process noise (full coverage, 66 steps). Blue: $\lambda_{\max}(P_i)$ below the gate; purple: exceeded. Cyan: 2σ covariance. Red: certified.

***Abstract*— Conventional coverage path planning considers a region covered once it intersects a sensor's geometric footprint. However, this assumption fails in decentralized multi-robot missions operating within GPS-denied environments. Because each robot relies on dead reckoning from a noisy inertial measurement unit (IMU), geometric visits alone cannot guarantee that pose uncertainty is sufficiently bounded for verification. To ensure certified coverage, an explicit constraint is required: the maximum eigenvalue of an agent's belief covariance matrix, $\lambda_{\max}(P_i)$, must remain below a designated threshold. When robots enter mutual communication range, inter-robot range measurements can be leveraged to reduce state estimation uncertainty. This introduces a coupled decision-making problem involving spatial coverage, exiting the GPS-denied zone to re-anchor against GPS, and maintaining proximity to GPS-anchored teammates. We formulate this task as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP) and train a shared policy using a convolutional network operating on a fixed-size egocentric local observation. Under severe inertial drift, the unprivileged policy outperforms privileged classical planners, including receding-horizon controllers that forecast covariance and range. Inter-robot ranging is required in the onboard localization filter, not as a policy feature. Frozen weights transfer zero-shot across map scale and team size at the default 7×7 crop and the nominal certification bound used in training; transfer degrades under a smaller crop or when that bound is tightened.**

***Index Terms*— coverage planning, GPS-denied navigation, pose uncertainty, multi-agent reinforcement learning, cooperative localization**

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I. Introduction

Coverage path planning is a foundational problem in multi-robot exploration, inspection, and mapping [1], [2]. Most classical planners assume that a spatial region is fully observed once it falls within a sensor's geometric field of view. However, this assumption breaks down in GPS-denied environments, such as subterranean tunnels, urban canyons, and indoor structures. In these settings, robots dead-reckon using noisy inertial measurement units (IMUs), causing the position covariance P_i to grow continuously over time. Consequently, a trajectory that appears complete on a nominal map may leave large portions of the environment uncertified if the maximum eigenvalue of that covariance, $\lambda_{\max}(P_i)$, exceeds a required safety threshold. The primary objective is therefore not merely spatial traversal, but ensuring that $\lambda_{\max}(P_i)$ remains bounded below the covariance threshold while the team completes grid coverage. Fig. 1 shows one greedy trial under moderate process noise.

This problem formulation introduces three main challenges. First, the planner must reason explicitly about *certified* coverage rather than pure spatial occupancy: a visit counts only when the agent is sufficiently certain of its own position, which we enforce by gating coverage on the agent's pose covariance (formalized later as (8)). Second, the decision space couples three interdependent objectives: exploring unvisited locations, exiting the GPS-denied zone to

Manuscript 7158 submitted to 2027 IEEE International Conference on Robotics and Automation (ICRA). Received September 15, 2026.

reset P_i via GPS re-anchoring, and remaining within range of teammates so that cooperative range updates can bound P_i. An agent that spends excessive time in distant regions accumulates unbounded drift and fails to certify observations, whereas an agent that re-anchors too frequently reduces team coverage efficiency. Third, the observation space must remain invariant to overall map dimensions to support scale-independent zero-shot policy transfer.

We model these coupled decisions as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP) [3] and learn a shared convolutional policy that receives a 7×7 four-channel egocentric observation window. An onboard cooperative range extended Kalman filter (EKF) handles localization. Coverage certification and rewards are evaluated using $\lambda_{\max}(P_i)$, whereas the actor network receives the scalar matrix trace $\mathrm{tr}(P_i)$ as an input. Policy optimization is conducted using Multi-Agent PPO (MAPPO) [4]. Model performance is evaluated using a frozen checkpoint trained on a 10×10 map and evaluated greedily at matched cell resolution (w=0.1) on 10×10, 20×20, and 30×30 grids, at the nominal certification bound used in training and under a tightened dead-reckoning stress test.

This paper presents three principal contributions, validated in Section V and Appendix III:

1) Under high inertial drift, the proposed local policy achieves higher coverage than privileged classical controllers operating under the same certification predicate, including receding-horizon planners that forecast covariance growth and range to GPS-anchored agents (Section V-A).
2) Frozen network weights exhibit zero-shot generalization across map scales and team sizes at the default crop and the nominal certification bound used in training. Transfer degrades when the crop is reduced or when that bound is tightened below the dead-reckoning budget of a larger map (Section V-B).
3) An extensive multi-seed evaluation and ablation suite establishes robustness of the approach: ranging lesions show that cooperative range updates matter inside the EKF rather than as policy features (Section V-C); architecture and MARL-algorithm ablations isolate the egocentric crop and confirm that Independent PPO matches MAPPO while QMIX does not; and matched grid versus turn-speed studies show that waypoint commands remain a practical outer loop under more realistic motion constraints (Appendix III).

II. Related Work

A. Belief-Space Planning and Uncertainty-Aware Coverage

Standard coverage path planners—including Voronoi partitioning [5], cell decomposition [1], [2], and next-best-view approaches [6]—assume perfect localization and evaluate coverage purely through geometric visibility. In unmapped or GPS-denied environments, belief-space planning incorporates state-estimation uncertainty into decision-making [7]. However, trajectory optimization in belief space is computationally intensive, as predicting continuous nonlinear covariance dynamics scales poorly with team size and horizon length. To benchmark our work against these paradigms, we evaluate against privileged classical planners operating under the same uncertainty constraint, including receding-horizon Model Predictive Control (MPC) baselines that explicitly forecast belief covariance $\lambda_{\max}(P_i)$ and range dynamics to GPS-anchored teammates. Our approach offers a lightweight, learning-based decentralized policy that replaces expensive online belief-space trajectory optimization.

B. Learned Active Exploration and Scale Transfer

Deep reinforcement learning (DRL) has shown significant promise for multi-robot spatial exploration. ARiADNE [8], [9] and MARVEL [10] transfer local neural exploration policies across map scale and team size; IR^2 [11] addresses exploration under sparse intermittent connectivity. These methods plan over occupancy or graph features under uncorrupted state estimates and do not model localization drift. We extend scale-invariant multi-agent reinforcement learning (MARL) to joint spatial coverage and belief-space dynamics. Crucially, we show that zero-shot scale transfer under pose uncertainty is not merely a function of spatial visibility, but is fundamentally bounded by the physical growth of position covariance relative to the certification gate threshold c_{cov}.

C. Cooperative Localization

Inter-robot ranging bounds pose error in GPS-denied multi-robot systems whenever a teammate retains absolute state information [12], [13]. As detailed in Section III-C, we apply a unified scalar range update across two operating regimes. When the ranging partner operates outside the GPS-denied zone, the update is absolute, treating the partner's position as a deterministic anchor with zero uncertainty. Conversely, when both agents reside within the denied zone, the update incorporates the partner's full belief distribution—factoring in its covariance to prevent overconfident updates from uncertain sources. Consequently, the learned policy coordinates area coverage while periodically driving agents to exit the denied zone to re-anchor and maintain inter-robot proximity, seeking to keep pose uncertainty below the certification threshold.

III. Problem Formulation

A. Environment

The workspace is an obstacle-free plane. Agent i has planar pose $(p_i, \theta_i) \in \mathbb{R}^2 \times S^1$, believed position $\hat{p}_i \in \mathbb{R}^2$, and covariance $P_i \in \mathbb{R}^{2\times2}$. A square GPS-denied zone $\mathcal{Z} = [-z, z]^2$ is centered at the origin. GPS is available the instant p_i leaves $\mathcal{Z}$: we assume a perfect fix, so $\hat{p}_i \leftarrow p_i$ and the covariance is reset to $P_i \leftarrow 0$ on that step, with no residual uncertainty and no IMU process noise. Inside $\mathcal{Z}$, the agent dead-reckons from a noisy IMU and P_i grows. A team of N agents spawns just outside $\mathcal{Z}$. Coverage targets are the $(n+1)^2$ vertices $\mathcal{P}$ of an $n \times n$ lattice of cell width $w = 2z/n$ that tiles $\mathcal{Z}$.

Manuscript 7158 submitted to 2027 IEEE International Conference on Robotics and Automation (ICRA). Received September 15, 2026.

B. Motion Models

Each control interval of duration Δt is a rest-to-rest bang-bang hop whose IMU noise is derived in Section III-C. We compare two motion models that share this hop, the same process-noise map $Q_i(v,\omega)$, and the same certification predicate. Grid waypoints are the primary model used in the main experiments; the turn-speed unicycle is a more realistic alternative with yaw and speed constraints.

Grid waypoints. At each step the agent may stay or take a cardinal hop of length $v_0\Delta t$ along $(\pm x, \pm y)$. Heading snaps to the travel direction, so a covering step is already aligned with the forward half-plane of the coverage sensor. The filter is evaluated at $\omega{=}0$ and speed v_0 (zero on stay), so the process-noise covariance Q_i has no centripetal (cross-track) contribution.

Turn-speed unicycle. At each step the agent commands a pair (ω, v) from $\omega \in \{-\omega_{\max}, 0, +\omega_{\max}\}$ and $v \in \{0.5, 1.0, 1.5\}v_0$ (nine choices). Pose integrates as a unicycle on the plane. Q_i uses the commanded (v,ω): yaw increases cross-track IMU noise and a $1.5\times$ hop inflates along-track noise (both standard deviations are defined with the process-noise model in Section III-C). Extra yaw steps are often required before a lattice corner enters the heading window, and those steps accumulate P_i before certification can occur.

C. Onboard Filter

The onboard estimator is a position-only EKF: the state is $(\hat{p}_i, P_i)$. Heading θ_i is the integrated command and is not estimated, so the process Jacobian is $F = I_2$. True pose integrates noiseless kinematics. Inside $\mathcal{Z}$ the estimate is a dead-reckoning copy of the same hop with additive IMU noise; P_i is the covariance of that copy.

Each control interval of duration Δt is a rest-to-rest bang-bang hop (Section III-B): a triangular speed profile that covers path length $|v|\Delta t$ from rest. Peak speed is $2|v|$, so the tangential acceleration, the centripetal acceleration at peak speed, and the reference hop at nominal speed v_0 with $\omega = 0$ are

$$a_t = \frac{4|v|}{\Delta t}, \qquad a_n = 2|v|\,|\omega|, \qquad a_{\mathrm{ref}} = \frac{4v_0}{\Delta t}. \tag{1}$$

Along-track and cross-track IMU standard deviations, $\sigma_\parallel$ and $\sigma_\perp$, scale with those accelerations,

$$\sigma_\parallel = p_n \max\!\big(a_t/a_{\mathrm{ref}},\, \epsilon\big), \qquad \sigma_\perp = \tfrac{1}{2}\sigma_\parallel + p_n\,(a_n/a_{\mathrm{ref}}), \tag{2}$$

where p_n is the inertial drift coefficient and $\epsilon = 0.1$ is a stationary floor (residual on stay). With body-frame unit vectors $u = (\cos\theta_i, \sin\theta_i)$ and $n_\perp = (-\sin\theta_i, \cos\theta_i)$,

$$Q_i = \sigma_\parallel^2\, uu^\top + \sigma_\perp^2\, n_\perp n_\perp^\top. \tag{3}$$

The same Q_i is used both to sample the IMU residual and to inflate P_i. On a grid hop, $\omega = 0$ and $v \in \{0, v_0\}$, so $a_n = 0$ and the cross-track term collapses to $\sigma_\perp = \frac{1}{2}\sigma_\parallel$. On a turn-speed hop the commanded (v,ω) enters Q_i directly.

a) Predict: Let Δp_i be the kinematic displacement of the hop. If the updated position satisfies $p_i \in \mathcal{Z}$,

$$\hat{p}_i \leftarrow \hat{p}_i + \Delta p_i + w_i, \qquad P_i \leftarrow P_i + Q_i, \qquad w_i \sim \mathcal{N}(0, Q_i). \tag{4}$$

If $p_i \notin \mathcal{Z}$, GPS is recovered on that step: $\hat{p}_i \leftarrow p_i$, $P_i \leftarrow 0$, and Q_i is not applied.

b) Range update: A scalar range is obtained when $\|p_i - p_j\|_2 \le r_c$ and receiver i is inside $\mathcal{Z}$. The measurement model is

$$z_{ij} = \|p_i - p_j\|_2 + \nu, \qquad \nu \sim \mathcal{N}(0, R), \qquad R = 10^{-4}. \tag{5}$$

The partner treatment depends on GPS. If $p_j \notin \mathcal{Z}$, the partner is a GPS anchor: the predicted range uses the true position p_j with $P_j{=}0$ (absolute update). If both agents are inside $\mathcal{Z}$, the predicted range uses the partner belief $\hat{p}_j$ and the partner covariance P_j enters the innovation covariance, so an uncertain source cannot overconfidently shrink P_i. Linearizing about $\hat{p}_i$ with partner location $p_j^{\mathrm{ref}} \in \{p_j, \hat{p}_j\}$ gives $\hat{h} = \|\hat{p}_i - p_j^{\mathrm{ref}}\|_2$ and Jacobians

$$H_i = \frac{(\hat{p}_i - p_j^{\mathrm{ref}})^\top}{\hat{h}}, \qquad H_j = -H_i \in \mathbb{R}^{1\times 2}. \tag{6}$$

The scalar Kalman update is

$$\begin{aligned} S &= H_i P_i H_i^\top + H_j P_j H_j^\top + R, \\ L &= P_i H_i^\top S^{-1}, \\ \hat{p}_i &\leftarrow \hat{p}_i + L(z_{ij} - \hat{h}), \\ P_i &\leftarrow (I_2 - L H_i)\, P_i, \end{aligned} \tag{7}$$

with $P_j{=}0$ for a GPS anchor. The update is applied only if the residual passes a 3σ innovation gate, $|z_{ij} - \hat{h}| \le 3\sqrt{S}$. Under the linear-Gaussian model the residual is $\mathcal{N}(0, S)$, so this is the scalar Mahalanobis test $(z_{ij} - \hat{h})^2/S \le 9$ and rejects outliers or linearization failures. The gate does not change the Kalman gain. Each in-range teammate yields one such update, applied sequentially.

D. Sensor Footprints, Certification, and Reward

Three geometric footprints are used, all in the plane. There is no camera image or occupancy map; orientation is a directional predicate on the coverage disk.

Coverage. Lattice corner $c \in \mathcal{P}$ lies in agent i's coverage footprint when it is inside a disk of radius r_c *and* in the forward half-plane of θ_i. That disk is the sensor's geometric footprint in the sense of classical coverage planning. Certification additionally requires the covariance gate:

$$\begin{gathered} \|p_i - c\| < r_c, \quad \lambda_{\max}(P_i) < c_{\mathrm{cov}}, \\ |\theta_i - \angle(c - p_i)| < \pi/2. \end{gathered} \tag{8}$$

A geometric visit that fails the gate does not count. Once certified, a corner remains covered. Training uses the nominal bound $c_{\mathrm{cov}}{=}0.0125$; a tightened bound $c_{\mathrm{cov}}{=}10^{-4}$ appears only in the scale-transfer stress test of Section V-B.

Ranging and communication. Agents i and j share a scalar range if $\|p_i - p_j\|_2 \le r_c$ (omnidirectional disk, same radius as

coverage). Every in-range pair with receiver i inside $\mathcal{Z}$ enters the EKF via (7): absolute when $p_j \notin \mathcal{Z}$ (P_j=0), otherwise against $(\hat{p}_j, P_j)$.

GPS. A binary footprint. The moment an agent leaves $\mathcal{Z}$, covariance is assumed to reset to $P_i = 0$ and the belief snaps to true position. There is no GPS inside $\mathcal{Z}$.

State uncertainty that restricts certification is $\lambda_{\max}(P_i)$. The step reward for agent i is

$$r_i = -0.05 + \sum_{c \in \mathcal{C}_i} \frac{w_{\text{cover}}}{1 + \lambda_{\max}(P_i)}, \tag{9}$$

where $\mathcal{C}_i$ is the set of lattice corners newly certified by agent i on the current step. The two terms align with the objective as follows. The constant -0.05 is a per-step living cost: it raises return for finishing the lattice sooner, so the policy is pressured to certify quickly rather than linger until horizon T. The sum is the primary task signal: each newly certified corner pays $w_{\text{cover}}/(1+\lambda_{\max}(P_i))$. The denominator down-weights credits earned under high pose uncertainty, coupling spatial progress to the same covariance gate that defines certified coverage. The ego observation vector still provides matrix trace $\operatorname{tr}(P_i)$ as a compact scalar feature of the 2×2 covariance, distinct from the certification gate metric $\lambda_{\max}(P_i)$.

An episode terminates when all lattice corners are certified or maximum horizon T is reached.

IV. Method

A. Observations and Actions

The actor and the centralized critic consume raster cell masks on the $n \times n$ lattice that tiles $\mathcal{Z}$, not camera images (Fig. 2). A cell is marked certified on these maps only when all four of its lattice corners satisfy (8); otherwise it is uncertified. The two motion models of Section III-B share this observation, the reward (9), the filter of Section III-C, and the gate (8). Grid waypoints are the primary model; turn-speed is the more realistic unicycle alternative.

a) Actor crop: The policy input is an ego feature vector concatenated with a four-channel $K \times K$ crop, K=7 by default, centered on agent i's *believed* cell. Channel 0 is uncertified cells, channel 1 certified cells, channel 2 in-range teammates (a 1 at the believed cell of j when $\|p_i - p_j\|_2 \le r_c$), and channel 3 a binary out-of-bounds mask for crop cells that fall outside the episode lattice. Because K is fixed, the convolutional trunk does not depend on the global map size n.

The ego vector has length $8+3(N-1)$: heading, $\hat{p}_i$, $\operatorname{tr}(P_i)$, and a binary flag that $\hat{p}_i$ lies in $\mathcal{Z}$; then, for each other agent, the triple (true range, $\operatorname{tr}(P_j)$, unsigned bearing of $\hat{p}_j$ relative to θ_i), replaced by $(-1,-1,-1)$ beyond r_c; and finally the vector from $\hat{p}_i$ to the uncertified-corner centroid and the global certified fraction.

b) Critic map: MAPPO's critic is privileged [4]. Its global vector stacks $(\theta_i, p_i, \hat{p}_i, \operatorname{tr}(P_i))$ for every agent. The accompanying four-channel $n \times n$ map uses true positions: uncertified, certified, occupancy, and a covariance heat

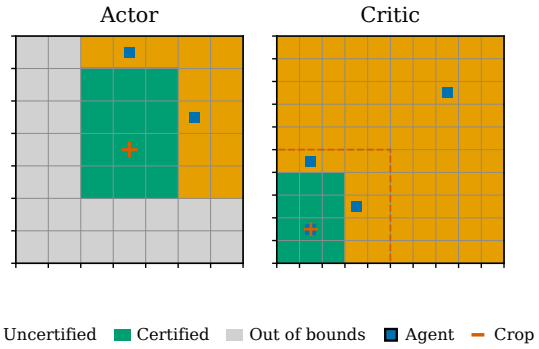


Fig. 2. Actor 7×7 crop and critic $n \times n$ map, same colors. Orange / green: uncertified / certified. Gray: out of bounds (actor only). Blue square: agent cell (actor: in-range $\hat{p}_j$; critic: true p_i). Dashed box: crop. Orange cross: ego.

$\operatorname{clip}(\lambda_{\max}(P_i)/c_{\text{cov}}, 0, 1)$ written at each occupied cell. The critic map grows with n; adaptive average pooling keeps the parameter count fixed.

Grid actions are stay and the four cardinal hops. Turn-speed uses the nine (v, ω) pairs of Section III-B.

B. Network Architecture and Training

The crop feature grid is processed by two 3×3 convolutional layers (4→32, 32→64, ReLU activation). Extracted spatial features are concatenated with the ego vector through a 256-unit dense layer, followed by two 512-unit tanh layers generating categorical action distributions. Neural network weights are shared across all team agents. Policy optimization utilizes MAPPO [4] with Generalized Advantage Estimation (GAE) [14] implemented in JAX [15]. Training runs for 10^7 environment steps across eight parallel environments, using rollout horizon 128, discount factor γ=0.99, GAE coefficient λ=0.95, PPO clipping parameter 0.1, entropy regularizer 0.01, and linearly annealed learning rate 3×10^{-4}. Inertial drift parameter p_n is resampled per episode from $\{5\times10^{-4}, 0.03, 0.1, 0.2\}$. For multi-size training, map half-extent z is sampled from $\{0.5, 1.0, 1.5\}$ at cell width w=0.1. Complete hyperparameter settings are provided in Appendix I.

Fig. 3 shows the mixed-noise grid run used for the reported checkpoint. Policy entropy falls from near-uniform $\ln 5$ to about 0.5 nats (perplexity ≈1.6 on five actions). Over the same 10^7 steps the Fiedler value λ_2 of the communication-range graph [16] rises from ≈0.19 to ≈0.48. That rise is not rewarded explicitly; it is a side effect of learning to keep a ranging path to a GPS-anchored teammate so that (7) can hold $\lambda_{\max}(P_i)$ below the gate. The same run shows that training jointly optimizes certified coverage and time. Mean on-policy episode length falls from ≈187 to ≈112 steps as coverage rises from 0.72 to 0.99, and keeps falling after coverage saturates: the per-step cost in (9) makes shorter missions higher return, so MAPPO compresses trajectories rather than lingering at the horizon. Because P_i grows with time inside $\mathcal{Z}$, finishing sooner also leaves less interval for inertial drift. These lengths mix p_n and ε-greedy noise;

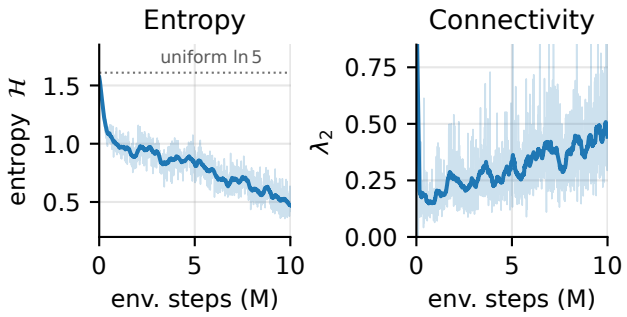


Fig. 3. Mixed-noise MAPPO training (10^7 steps). Left: policy entropy (thin: raw; thick: smoothed). Dotted: uniform $\ln 5$. Right: algebraic connectivity λ_2 of the r_c range graph.

greedy default evaluation finishes in a median of 77 steps (83% complete before T=400).

V. Experiments

Unless otherwise stated, policies are evaluated greedily on a 10×10 lattice with cell width w=0.1 at the nominal certification bound c_{cov}=0.0125 used in training. Comparisons against classical planners use five independent training seeds and twenty episodes per seed. Scale-transfer experiments use four seeds at matched drift p_n=0.002 and, where noted, a tightened bound $c_{\mathrm{cov}}=10^{-4}$. We report the sample mean and a parametric 95% confidence interval, and assess pairwise differences with Welch's t-test and the Mann–Whitney U test.

A. Comparison with Classical Planners

All classical baselines receive privileged state: true pose p_i, covariance P_i, and the global uncertified set. Nearest-greedy selects the closest uncertified corner and leaves $\mathcal{Z}$ when $\lambda_{\max}(P_i)$ exceeds half the certification threshold. Independent receding-horizon MPC enumerates cardinal actions and scores them by predicted process-noise growth and range to a GPS-anchored teammate. Joint receding-horizon MPC applies the same model as a Gauss–Seidel best response over the team. The dedicated-beacon, lawnmower, and Voronoi controllers obey the same nominal gate (8) and the same GPS-reset rule.

Table I and Fig. 4 report certified coverage versus inertial drift under that nominal bound. At low p_n, every closed-loop controller, including the learned policy, saturates. As drift increases, the unprivileged policy continues to certify a large fraction of the lattice, whereas nearest-greedy, both receding-horizon controllers, and the dedicated-beacon planner degrade sharply; the open-loop lawnmower and Voronoi controllers degrade earlier. Those planners therefore fail under the same certification predicate used to train the policy. Extending the episode horizon does not close the gap: privileged access to P_i and a short-horizon covariance forecast are insufficient, at high drift, to match a policy trained to jointly cover and re-anchor.

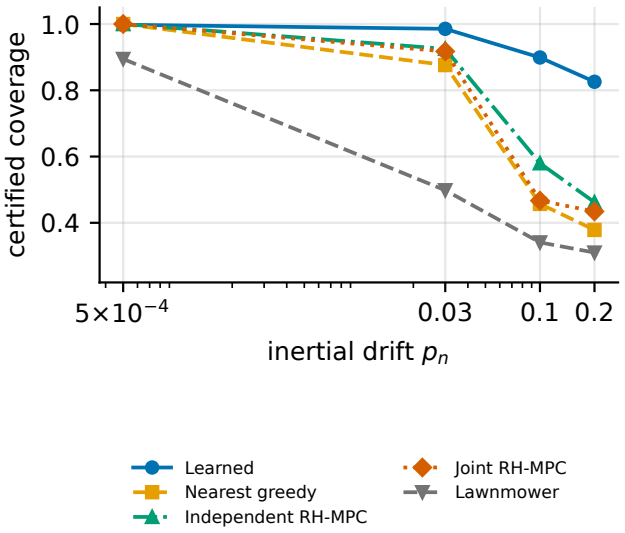


Fig. 4. Certified coverage versus inertial drift under the nominal gate (Table I). The learned policy is unprivileged; all other methods observe true pose and P_i.

B. Zero-Shot Transfer

Zero-shot transfer across map size is limited by the certification bound rather than by the convolutional crop alone. Table II compares a checkpoint trained only on 10×10 maps with an otherwise identical policy trained on randomized map sizes, both evaluated at matched cell width. At the nominal bound c_{cov}=0.0125, the frozen 10×10 weights retain near-complete coverage on 20×20 and 30×30 lattices that never appear in that training distribution, and they match or exceed the size-randomized policy on the largest map. When the bound is tightened to $c_{\mathrm{cov}}=10^{-4}$, the same frozen weights lose coverage as n increases, while size-randomized training retains substantially more. At fixed w, a larger lattice requires a longer excursion before a GPS reset, so $\lambda_{\max}(P_i)$ exceeds the reduced bound independently of whether the observation window is scale-invariant.

TABLE II

Map-size transfer at the nominal and tightened certification bounds (mean and 95% CI; four training seeds, matched drift p_n=0.002).

Gate	Training maps	10×10	20×20	30×30
Nominal	10×10 only	1.000 [1.000, 1.000]	0.999 [0.998, 1.001]	0.996 [0.990, 1.001]
	multi-size	1.000 [1.000, 1.000]	0.995 [0.985, 1.005]	0.867 [0.716, 1.018]
Tightened	10×10 only	0.773 [0.630, 0.916]	0.277 [0.231, 0.323]	0.125 [0.121, 0.128]
	multi-size	0.998 [0.994, 1.002]	0.843 [0.771, 0.915]	0.531 [0.442, 0.619]

Table III varies team size N and crop width K on the same frozen 10×10 checkpoint. With K=7, teams of three or more agents transfer reliably; N=2 remains high on average but with large episode variance, consistent with having only one possible GPS-anchored ranging partner. Reducing the crop to K=5 degrades coverage, whereas increasing it to K=11 yields no further gain.

TABLE I

GREEDY COVERAGE VS INERTIAL DRIFT (MEAN, 95% CI). LEARNED: FIVE SEEDS, TWENTY EPISODES. BASELINES ARE PRIVILEGED. LAWNMOWER AND VORONOI USE T=200; OTHERS T=400. INDEPENDENT RH-MPC: H=6; JOINT: H=4.

Method	default	p_n=0.03	p_n=0.1	p_n=0.2
Learned policy	0.998 [0.997, 0.999]	0.985 [0.977, 0.994]	0.899 [0.883, 0.915]	0.826 [0.808, 0.843]
Nearest greedy	1.000 [1.000, 1.000]	0.876 [0.854, 0.898]	0.457 [0.438, 0.476]	0.379 [0.366, 0.391]
Independent RH-MPC	1.000 [1.000, 1.000]	0.925 [0.907, 0.943]	0.580 [0.554, 0.605]	0.462 [0.441, 0.483]
Joint RH-MPC	1.000 [1.000, 1.000]	0.917 [0.862, 0.972]	0.467 [0.396, 0.538]	0.434 [0.395, 0.474]
RH-MPC + GPS beacon	1.000 [1.000, 1.000]	0.956 [0.941, 0.972]	0.549 [0.524, 0.573]	0.460 [0.439, 0.480]
Lawnmower	0.895 [0.864, 0.925]	0.498 [0.475, 0.521]	0.341 [0.326, 0.356]	0.310 [0.296, 0.324]
Voronoi	0.959 [0.940, 0.977]	0.459 [0.439, 0.478]	0.300 [0.284, 0.316]	0.267 [0.252, 0.282]

TABLE III

ZERO-SHOT TEAM-SIZE AND CROP-SIZE TRANSFER FROM A FROZEN 10×10 CHECKPOINT (GREEDY MEAN ± STD; n=20 TEST EPISODES PER CELL).

N	K=5	K=7	K=11
2	0.669±0.269	0.955±0.198	0.955±0.197
3	0.661±0.236	1.000	0.999±0.002
5	0.916±0.104	1.000	1.000
8	0.883±0.122	1.000	1.000

Table IV replaces the convolutional trunk with an MLP on the same local crop and with an MLP on the full map. The crop MLP preserves transfer; the global MLP does not. The transferable representation is therefore the fixed-size egocentric window, not the particular convolutional parameterization.

TABLE IV

NETWORK ARCHITECTURE TRANSFER UNDER MATCHED CELL WIDTH (GREEDY MEAN ± STD).

Arch	Train	z=0.5	z=1.5
CNN	10×10 only	0.999±0.002	0.785±0.082
CNN	multi-size	0.997±0.006	0.614±0.134
MLP crop	10×10 only	1.000	0.765±0.065
MLP crop	multi-size	0.995±0.009	0.576±0.149
MLP global	10×10 only	0.976±0.033	0.327±0.062
MLP global	multi-size	0.889±0.208	0.294±0.143

C. Role of Cooperative Ranging

Table V holds the environment, the certification gate, and the observation layout fixed, and removes either the actor's range features or the EKF range update (7). At high drift, masking range in the policy input leaves certified coverage statistically indistinguishable from the intact controller, whereas disabling the filter update produces a large drop. At low drift both interventions are inactive, because process noise does not drive $\lambda_{\max}(P_i)$ through the gate. Cooperative ranging is therefore required inside the filter (7), not as a policy feature: absolute when the partner is GPS-anchored, and otherwise with the partner's covariance folded into S.

TABLE V

RANGE MASKING VERSUS EKF MEASUREMENT UPDATES (MEAN AND 95% CI; EIGHT MATCHED GREEDY EPISODES).

Intervention	default	p_n=0.1
Control	1.000 [1.000, 1.000]	0.956 [0.914, 0.997]
Mask actor range	1.000 [1.000, 1.000]	0.967 [0.944, 0.990]
EKF range off	1.000 [1.000, 1.000]	0.687 [0.617, 0.757]

The high-drift advantage over privileged planners is reproduced by Independent PPO and is not reproduced by QMIX [17] (Appendix III). Grid waypoints outperform a more realistic turn-speed unicycle under the same filter and gate; the waypoint commands remain a practical outer loop because a low-level controller can track them (Appendix III).

VI. CONCLUSION

GPS-denied multi-robot coverage requires bounding $\lambda_{\max}(P_i)$, not merely traversing the lattice. Under that predicate, and under high inertial drift, a local convolutional policy outperforms privileged classical controllers, including receding-horizon MPC that forecasts covariance growth and range to GPS-anchored teammates. Frozen weights transfer zero-shot across map scale and team size at the default crop and the nominal certification bound used in training; transfer degrades under a smaller crop or when that bound is tightened below the dead-reckoning budget of a larger map. Multi-seed ablations further support robustness: cooperative ranging is required in the EKF update and is dispensable as a policy input; the egocentric crop—not a global map MLP—carries transfer; and Independent PPO matches MAPPO while QMIX does not.

This study evaluates kinematic motion in obstacle-free environments with ideal disk communication and a simplified process-noise model. Future work is to extend the same certification problem to planning in 3-D with obstacles, and to validate the policy on real hardware.

ACKNOWLEDGMENT

Portions of this manuscript—including text, result processing scripts, and figures—were drafted or revised using Cursor. All AI-assisted content was reviewed, edited, and verified by the authors, who take full responsibility for the final paper.

APPENDIX I
TRAINING CONFIGURATION

Table VI provides standard environment constants and optimization hyperparameters. Drift rate p_n is resampled per episode from the four-value set. Training and default evaluation use the nominal bound $c_{\mathrm{cov}}{=}0.0125$; Table II additionally reports a tightened bound $c_{\mathrm{cov}}{=}10^{-4}$.

TABLE VI
ENVIRONMENT AND OPTIMIZATION HYPERPARAMETERS.

Parameter	Value
Agents N	5
Zone half-extent z	0.5
Cell width w	0.1
Nominal hop speed v_0	0.2
Corners $\lvert\mathcal{P}\rvert$	121 at $10{\times}10$
Communication / range radius r_c	0.3
Range measurement $R = r_{\mathrm{cov}}$	10^{-4}
Coverage reward w_{cover}	0.5
Covariance gate c_{cov}	0.0125 (nominal); 10^{-4} (tightened)
Stationary IMU floor ϵ	0.1
Drift mix p_n	$\{5{\times}10^{-4}, 0.03, 0.1, 0.2\}$
Step Δt	0.1
Episode horizon T	400
Total environment steps	10^7
Learning rate	3×10^{-4}, linearly annealed
Parallel environments	8
Rollout length	128
Discount γ / GAE λ	0.99 / 0.95
PPO clip / entropy coefficient	0.1 / 0.01
Actor / critic epochs	4 / 4

APPENDIX II
JOINT RECEDING-HORIZON PROTOCOL

Joint and Independent receding-horizon MPC share the grid, the gate, and the reset seeds (Table VII). Independent MPC with the longer horizon wins the paired high-drift trials. Joint Gauss–Seidel search over a shorter team horizon can replace a feasible exit with a locally higher H-step score.

TABLE VII
PAIRED COVERAGE COMPARISON ON IDENTICAL ENVIRONMENT RESETS (SEEDS 1000–1019). INDEPENDENT MPC USES $H{=}6$; JOINT MPC USES $H{=}4$. Δ REPRESENTS JOINT MINUS INDEPENDENT COVERAGE DIFFERENCE.

Scenario	Independent	Joint	Δ	Worse
default	1.000	1.000	$+0.000$	0
$p_n{=}0.03$	0.965 [0.954, 0.976]	0.917 [0.862, 0.972]	-0.048	8
$p_n{=}0.1$	0.571 [0.515, 0.627]	0.467 [0.396, 0.538]	-0.104	17
$p_n{=}0.2$	0.467 [0.414, 0.520]	0.434 [0.395, 0.474]	-0.033	10

APPENDIX III
ADDITIONAL ABLATIONS

Table VIII repeats training under the same budget with Independent PPO and QMIX [17]. IPPO matches MAPPO on the default and high-drift grids; QMIX does not. The high-drift result is therefore not an artifact of MAPPO's centralized critic.

TABLE VIII
MARL TRAINING ALGORITHM (GREEDY COVERAGE MEAN AND 95% CI).

Algorithm	default	$p_n{=}0.1$	coarse
MAPPO	0.997 [0.996, 0.998]	0.885 [0.866, 0.904]	0.685 [0.624, 0.746]
IPPO	0.998 [0.997, 0.999]	0.906 [0.897, 0.915]	0.543 [0.476, 0.609]
QMIX	0.850 [0.798, 0.902]	0.621 [0.578, 0.665]	0.267 [0.224, 0.309]

Table IX compares grid waypoints with the turn-speed unicycle, which imposes more realistic yaw and speed constraints. Grid coverage remains higher as p_n increases: under high drift, extra turning steps accumulate covariance before a lattice corner enters the coverage heading window. On coarse lattices ($w{=}0.2$) under low noise, turn-speed can cover multiple corners per heading change; that reversal does not hold at high drift. Even so, the weaker turn-speed result does not invalidate the grid policy: cardinal waypoints are an outer-loop interface that a low-level tracking controller can follow on a platform with continuous heading, so the learned waypoint sequence remains executable under more realistic motion constraints.

TABLE IX
GRID VERSUS TURN-SPEED COVERAGE UNDER MATCHED TRAINING (GREEDY, 100 EVALUATION EPISODES).

Scenario	Grid	TS
default	0.998 [0.997, 0.999]	0.831 [0.812, 0.849]
$p_n{=}0.03$	0.985 [0.977, 0.994]	0.833 [0.814, 0.851]
$p_n{=}0.1$	0.899 [0.883, 0.915]	0.363 [0.343, 0.382]
$p_n{=}0.2$	0.826 [0.808, 0.843]	0.305 [0.288, 0.323]
coarse ($5{\times}5$)	0.641 [0.573, 0.709]	0.721 [0.703, 0.739]
coarse, $p_n{=}0.1$	0.718 [0.650, 0.786]	0.416 [0.389, 0.443]
coarse, $p_n{=}0.2$	0.653 [0.590, 0.716]	0.350 [0.329, 0.372]
very coarse ($4{\times}4$)	0.670 [0.611, 0.730]	0.712 [0.691, 0.732]

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